

SWALIC: Asymmetric Wavelet Image Compression with Parallel Shift-Attention Architecture and Cross-Band Collaborative Refinement

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ABSTRACT

Real-time deployment of learned image compression (LIC) on resource-constrained sensors, such as UAVs, is hindered by high encoder complexity. We propose SWALIC, an asymmetric, wavelet-based lightweight LIC framework. Utilizing two-stage biorthogonal wavelet decomposition and a parallel attention-shift backbone, SWALIC employs a subband-aware complexity allocation strategy that prioritizes the second-level low-frequency subband. To preserve structural details with minimal overhead, a Cross-Band Modulation Module (CBMM) is integrated into the encoder to modulate high-frequency features. The decoder utilizes a progressive refinement scheme, comprising a Contextual Refinement Module (CRM) for feature restoration via Spatial Feature Transform (SFT) and a Post-Process Module for artifact mitigation. Evaluated on the UC Merced (UCM) dataset, SWALIC achieves rate-distortion performance comparable to advanced lightweight methods while drastically reducing encoder resources to only 0.119 M parameters and 0.044 GMACs, respectively. Our framework offers a favorable performance-complexity trade-off, demonstrating strong potential for real-time deployment on energy-constrained hardware.

1. Introduction

In the current era of ubiquitous earth observation, the proliferation of remote sensing platforms—ranging from sophisticated high-orbit satellites to agile unmanned aerial vehicles (UAVs)—has ushered in a new epoch of visual data explosion (Mohsan et al., 2022). These systems are now capable of capturing ultrahigh-resolution imagery with unprecedented temporal and spectral fidelity. However, this deluge of data has encountered a formidable bottleneck: the severe disparity between the massive volume of raw sensory data and the limited downlink bandwidth available for transmission (Yang et al., 2017). For edge-based platforms, low-latency image compression is no longer an auxiliary feature but an indispensable prerequisite for operational viability.

Furthermore, the design of on-board compression algorithms is strictly governed by the Size, Weight, and Power (SWaP) constraints inherent to aerospace and robotic hardware (Chintalapati et al., 2025; Sang et al., 2026). In these scenarios, a fundamental computational asymmetry defines the system architecture. On the sensor side (the encoder), energy reserves and processing cycles are at a premium, as excessive computation directly translates to reduced flight endurance or thermal instability. Conversely, the ground station (the decoder) typically possesses substantial computational resources, including high-performance GPUs and cloud-based infrastructures, capable of hosting complex reconstruction models (Wang et al., 2020). This intrinsic asymmetry creates an urgent demand for a paradigm shift in Learned Image Compression (LIC) research: we must transition from traditionally symmetric Variational Autoencoder (VAE) architectures toward radically asymmetric frameworks that prioritize extreme encoder-side efficiency while leveraging the decoder's capacity to restore high-fidelity details.

Considerable progress has been made in LIC over the past decade, and a comprehensive review of existing methods is provided in Section 2. In summary, while advanced LIC methods have surpassed traditional codecs in compression efficiency, their computational demands remain prohibitive for edge deployment. Recent lightweight designs, particularly those employing shift-based operations (Bao et al., 2025), have demonstrated that spatial shift can effectively replace heavy convolutions with minimal overhead. In parallel, frequency-domain decomposition methods (Mishra et al., 2021; Zhang et al., 2022; Ma et al., 2020) have shown promise, yet the potential of combining such decompositions with a radically asymmetric encoder remains largely uninvestigated.

Building on these insights, we introduce three targeted extensions that better suit our asymmetric wavelet-based framework. First, we replace the fixed hard shift with a *learnable* variant (LSS) that performs a weighted combination over multiple directions, enabling the shift pattern to adapt to the distinct spatial statistics of different wavelet subbands. Second, we augment the shift bottleneck with an inverted expand ratio (SBB), lifting features into a higher-dimensional space before spatial mixing for richer cross-channel interaction. Third, we reorganize the serial SSB+CRA arrangement into a *parallel, gated* structure (PASB), where channel attention directly modulates the shift output—a design particularly beneficial in our extremely low-dimensional encoders (16–48 channels) where every channel carries substantial information.

Consequently, even the current so-called "lightweight" models often exhibit a complexity-performance imbalance when deployed on resource-constrained hardware. To provide a holistic perspective on this challenge, we compare several advanced and lightweight LIC models across four critical dimensions in Fig. 1. As illustrated, while recent lightweight methods such as He24 (He et al., 2024) have notably reduced complexity, their encoder-side cost remains

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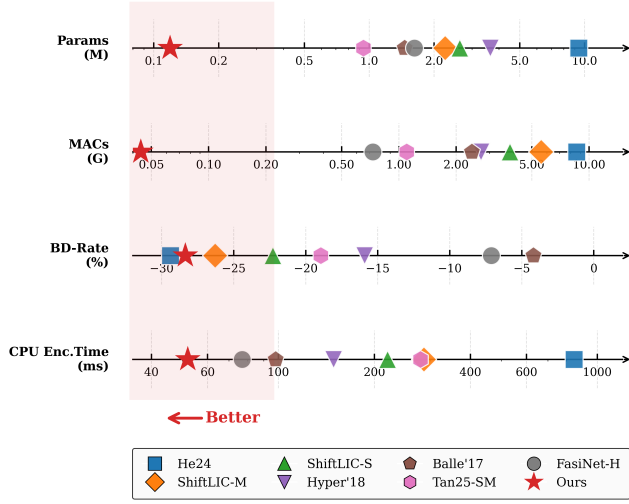


Figure 1: A comprehensive comparison of LIC performance across four dimensions. R-D performance is quantified using BD-Rate (%), which is anchored by JPEG2000. Our model (SWALIC) achieves an excellent trade-off between performance and complexity compared to representative works, such as He24 (He et al., 2024), ShiftLIC (Bao et al., 2025), Tan25 (Tan et al., 2025), and FasiNet (Zheng et al., 2021).

substantial for resource-constrained edge platforms. Conversely, although existing lightweight models reduce the overhead, they still occupy a sub-optimal region in terms of the trade-off between MACs, latency, and BD-Rate. This clear gap motivates the need for a radically asymmetric framework that can achieve advanced performance with "micro-scale" encoding complexity.

To bridge this gap, we propose SWALIC, a radically asymmetric wavelet-based LIC framework specifically tailored for real-time deployment on resource-constrained edge platforms. Central to our framework is a two-stage biorthogonal wavelet decomposition (*Bior4.4*) that orchestrates the input image into hierarchical subbands. This process effectively concentrates the core structural energy into the second-level low-frequency subband (LL_2) while isolating sparse, high-frequency textures into the residual components (HF_2 and HF_1).

Leveraging this frequency-domain decoupling, we implement a *subband-aware complexity allocation strategy*: a high-capacity network is dedicated to the information-rich LL_2 subband for high-fidelity base reconstruction, while the sparse high-frequency paths are processed by minimalist encoders. All backbones are built with the parallel attention-shift modules (LSS, SBB, and PASB) described above, replacing conventional convolutions throughout. This joint allocation of computational budget and shift behavior ensures that encoder-side resources are concentrated primarily where information density is highest.

Furthermore, SWALIC incorporates a dual-stage modulation and refinement mechanism to compensate for the potential fidelity loss inherent in lightweight encoding. At

the encoder side, we introduce the Cross-Band Modulation Module (CBMM), which exploits inter-subband structural correlations by utilizing low-frequency priors to pre-enhance high-frequency features before quantization, requiring negligible computational overhead. To restore fine-grained textures without increasing the on-board burden, a progressive refinement scheme is deployed at the decoder. This scheme integrates a Contextual Refinement Module (CRM) that executes Spatial Feature Transforms (SFT) for high-frequency restoration, followed by a PostProcess Module utilizing global residual learning to mitigate reconstruction artifacts. By strategically offloading the enhancement burden to the server-side decoder, SWALIC achieves a radical reduction in encoder complexity while maintaining competitive rate-distortion performance, as demonstrated in our empirical evaluations.

Our contributions are summarized as follows:

- (1) We propose SWALIC, a subband-aware asymmetric architecture that explicitly leverages the energy distribution of wavelet subbands. By assigning heterogeneous complexities to different frequency components, we achieve a significant reduction in encoder-side resources.
- (2) We introduce a lightweight, parallel attention-shift backbone with subband-specific learnable shifts. Comprising *Learnable Spatial Shift* (LSS), *Shift Bottleneck Block* (SBB), and *Parallel Attention-Shift Block* (PASB), this backbone replaces heavy convolutional layers with efficient shift-based operations and fuses channel attention with spatial shift in a parallel, gated structure.
- (3) We introduce a multi-stage enhancement scheme comprising the encoder-side CBMM and the decoder-side progressive restoration (consisting of CRM and PostProcess Module).
- (4) Extensive experiments demonstrate that SWALIC achieves R-D performance comparable to advanced lightweight methods while maintaining an extremely low encoder footprint of 0.119 M parameters and 0.044 GMACs.

2. Related Work

2.1. Traditional image compression

Traditional image compression algorithms follow a transform–quantization–entropy coding framework. JPEG (Wallace, 1991) uses the Discrete Cosine Transform (DCT) on 8×8 blocks, whose independent processing often produces blocking artifacts at low bitrates. JPEG2000 (Rabbani and Joshi, 2002) replaces the DCT with the Discrete Wavelet Transform (DWT) (Antonini et al., 1992), which decomposes the image into multiple frequency subbands through multiresolution analysis. By concentrating signal energy into a few low-frequency coefficients, DWT eliminates block artifacts and supports more flexible representations of image structures (Tausif et al., 2021). The energy compaction and multiresolution properties of DWT directly motivate our choice of wavelet decomposition as the front-end of SWALIC.

2.2. Learned image compression

The end-to-end LIC framework based on a variational autoencoder (VAE) architecture was pioneered by Ballé et al. (2017). This framework was subsequently extended with a hyperprior model (Ballé et al., 2018) that captures spatial dependencies in the latent representation through side information, and further with autoregressive context models (Minnen et al., 2018; He et al., 2021) that exploit causal or checkerboard-structured dependencies among latent elements. Channel-wise autoregressive priors (Minnen and Singh, 2020) and attention modules (Cheng et al., 2020) have been introduced to enhance the entropy model and the nonlinear transforms, respectively. More recent works continue to push the RD frontier with advanced architectures: ELIC (He et al., 2022) employs unevenly grouped space-channel context models, while He et al. (He et al., 2024) explore lightweight attention mechanisms for practical real-time compression. In a related direction, several works adopt wavelet or frequency-domain transforms within the learned framework. iWave (Ma et al., 2020) uses a CNN-based lifting scheme to learn wavelet-like transforms with perfect reconstruction, and its versatile extension iWave++ (Ma et al., 2022) unifies lossy and lossless compression in a single model. Rippel and Bourdev (Rippel and Bourdev, 2019) proposed a real-time adaptive compression system using wavelet-like decompositions. Despite these advances, the common trend across these methods is that improved RD performance comes with increasing model complexity, making them unsuitable for resource-constrained edge deployment where encoder computation must be minimized.

2.3. Lightweight learned image compression

The high computational cost of advanced LIC methods has motivated extensive efforts on model lightweighting, which can be broadly divided into two categories.

Post-hoc compression. One line of work applies compression techniques to already-trained models. Luo et al. (Luo et al., 2022) prune the hyperprior module for memory efficiency. Kim et al. (Kim et al., 2020) combine asymmetric autoencoder design with weight and filter pruning. Tan et al. (Tan et al., 2025) introduce structured sparsity learning with regularized pruning for efficient LIC. Sun et al. (Sun et al., 2025) propose Q-LIC, which quantizes latent representations through channel splitting. In the knowledge distillation direction, Chen et al. (Chen et al., 2025) distill a lightweight student from a high-capacity teacher, while Liu et al. (Liu et al., 2026) specifically target onboard remote sensing deployment with a distilled lightweight model.

Lightweight architecture design from scratch. Instead of compressing existing models, a second line of work designs inherently efficient architectures. FasiNet (Zheng et al., 2021) achieves real-time CPU compression using an asymmetric autoencoder with separable down/up-sampling modules. Tan et al. (Tan et al., 2025) propose a grouped transform for ultra-low-complexity LIC. Among these, ShiftLIC (Bao et al., 2025) makes a particularly influential observation: feature maps at different depths of an LIC backbone exhibit

substantial similarity, differing primarily in edge orientation and texture granularity rather than semantic content. Building on this insight, ShiftLIC replaces large-kernel convolutions with a 1×1 Conv–Shift– 1×1 Conv bottleneck (Spatial Shift Block, SSB) and supplements it with channel attention (CRA), achieving significant reductions in parameters and FLOPs while maintaining competitive RD performance. Our proposed backbone directly builds upon and extends this shift-based paradigm with subband-specific learnable shifts, inverted bottleneck expansion, and parallel gated attention–shift fusion.

2.4. Frequency-domain decomposition for LIC

Beyond the lightweighting efforts described above, a complementary line of work leverages frequency-domain decomposition to improve compression. Mishra et al. (Mishra et al., 2021) proposed WDAED, which uses a single-level Haar DWT to decompose the image into four subbands, each processed by a standard convolutional autoencoder with a deep super-resolution decoder. The four subbands are trained in parallel, but the encoder and decoder share a symmetric structure with no complexity differentiation across subbands. Zhang et al. (Zhang et al., 2022) introduced an interpretable framework with a learned frequency-pyramid decomposition that splits the latent representation into low-, middle-, and high-frequency components, each with independent probability estimation models. By aligning with the human visual system, this design achieves strong MS-SSIM performance, but does not address encoder complexity reduction. These methods collectively demonstrate the potential of frequency-domain processing for LIC, yet the combination of a fixed-wavelet decomposition with a radically asymmetric encoder–decoder design and subband-aware complexity allocation has not been explored in prior work.

3. Proposed Method

3.1. Overall architecture

The overall architecture of the proposed SWALIC is illustrated in Fig. 2. Our framework is designed as a radically asymmetric VAE specifically optimized for the strict SWaP constraints of edge sensing platforms. The core philosophy is to leverage the energy compaction property of the wavelet domain to achieve subband-aware complexity decoupling, thereby shifting the heavy computational burden of feature refinement to the server-side decoder.

Two-stage wavelet decomposition. The front-end of the encoder performs a two-stage 2-D Discrete Wavelet Transform (DWT) to decompose the input image $\mathbf{x} \in \mathbb{R}^{H \times W \times 3}$ into a hierarchical frequency representation. Formally, the first stage decomposes $\mathbf{x}$ into four subbands:

$$\{\mathbf{LL}_1, \mathbf{LH}_1, \mathbf{HL}_1, \mathbf{HH}_1\} = \text{DWT}(\mathbf{x}) \quad (1)$$

Subsequently, the first-level low-frequency subband $\mathbf{LL}_1$ is further decomposed:

$$\{\mathbf{LL}_2, \mathbf{LH}_2, \mathbf{HL}_2, \mathbf{HH}_2\} = \text{DWT}(\mathbf{LL}_1) \quad (2)$$

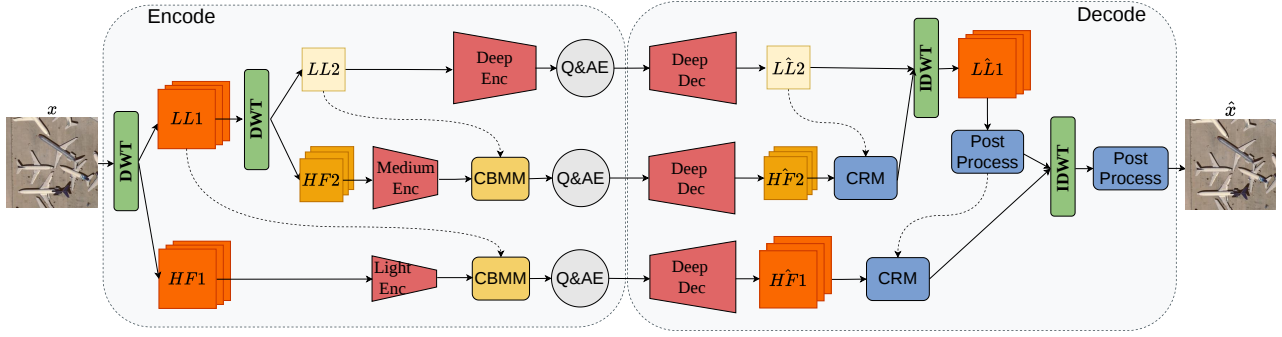


Figure 2: Overall architecture of the proposed SWALIC framework. The input image is decomposed into hierarchical subbands (LL_2, HF_2, HF_1) via a two-stage discrete wavelet transform (DWT). To minimize encoder-side resource consumption, subband-aware networks with heterogeneous complexities are employed: LL_2 is processed by a high-capacity network, whereas HF_2 and HF_1 utilize minimalist encoders enhanced by the Cross-Band Modulation Module (CBMM). All encoder and decoder backbones are built using parallel attention-shift modules—Parallel Attention-Shift Block (PASB) and Shift Bottleneck Block (SBB), each with subband-specific Learnable Spatial Shift (LSS). On the decoder side, a progressive refinement scheme, integrating Contextual Refinement Modules (CRM) and a multistage PostProcess Module, is applied to restore high-frequency details.

We specifically employ the *Bior4.4* wavelet for this process. Unlike orthogonal wavelets, biorthogonal wavelets provide a linear phase response, which is beneficial for preserving the spatial alignment of structural primitives during the reconstruction process (Cohen et al., 1992). Moreover, *Bior4.4* exhibits superior energy compaction, effectively concentrating the low-frequency global content into LL_2 while isolating high-frequency details into the sparse components $HF_2 = \{LH_2, HL_2, HH_2\}$ and $HF_1 = \{LH_1, HL_1, HH_1\}$.

Shift-based lightweight backbone. To construct a lightweight yet effective encoder-decoder backbone, we draw inspiration from recent findings in lightweight LIC design. As revealed in Bao et al. (2025), the feature maps extracted at different depths of an LIC backbone often exhibit substantial similarity—their primary differences lie in edge orientation and texture granularity rather than semantic content. Since edge information can be synthesized through zero-parameter spatial shift operations, replacing large-kernel convolutions with shift-based blocks offers a promising route to reduce computational redundancy. Additionally, following the separable convolution design of FasiNet (Zheng et al., 2021), our convolutional baseline factorizes standard $k \times k$ kernels (stride d) into $k \times 1$ (stride $d \times 1$) and $1 \times k$ (stride $1 \times d$) pairs, providing an already-efficient reference point against which the shift-based variants are evaluated.

Building on these insights, we redesign the encoder and decoder backbones using three shift-based building blocks: (i) a *Learnable Spatial Shift* (LSS) that replaces fixed hard shifts with a weighted combination of multiple shift directions; (ii) a *Shift Bottleneck Block* (SBB) that inherits the 1×1 Conv–Shift– 1×1 Conv bottleneck philosophy while adding channel expansion and GELU activation; and (iii) a *Parallel Attention-Shift Block* (PASB) that fuses channel attention and spatial shift operations in a parallel, gated interaction structure. Fig. 3 provides a side-by-side architectural comparison of the two backbones. As validated in Fig. 8 and

Table 4, the proposed shift-based backbone delivers both lower encoder complexity and improved R-D performance over its convolutional counterpart. The detailed designs of these modules are presented in Section 3.2.

Asymmetric subband-aware processing. Based on the multiresolution analysis, we implement a *subband-aware complexity allocation* strategy to minimize encoder-side resource consumption. The encoding process is partitioned into three parallel paths with heterogeneous capacities:

- (1) **Low-Frequency Path (LL_2):** Since LL_2 contains the critical structural backbone, it is processed by a high-capacity encoder E_{LL} with a larger channel budget and higher expand ratios in the shift blocks to ensure high-fidelity base reconstruction.
- (2) **High-Frequency Paths (HF_2, HF_1):** Given the sparsity of high-frequency components, we utilize extremely lightweight encoders E_{HF} with progressively reduced channel configurations.

The specific channel dimensions are provided in Fig. 3.

Crucially, the subband-aware design extends beyond channel allocation to the shift operations themselves: each subband maintains its own independent *Learnable Spatial Shift* (LSS) parameters, allowing the learnable shift weights to adapt to the distinct spatial statistics of low-frequency structures, mid-frequency textures, and high-frequency details respectively. In this way, not only “how many” channels are allocated per subband, but also “how” features are extracted—via subband-specific shift patterns—is tailored to each subband.

The motivation for adopting a wavelet-based decomposition goes beyond energy compaction alone. First, DWT acts as a fixed, computationally inexpensive front-end that already performs a substantial portion of decorrelation—the *Bior4.4* filter bank provides near-orthogonal frequency partitioning with linear phase, so the learned transforms start from a structured intermediate representation rather than raw

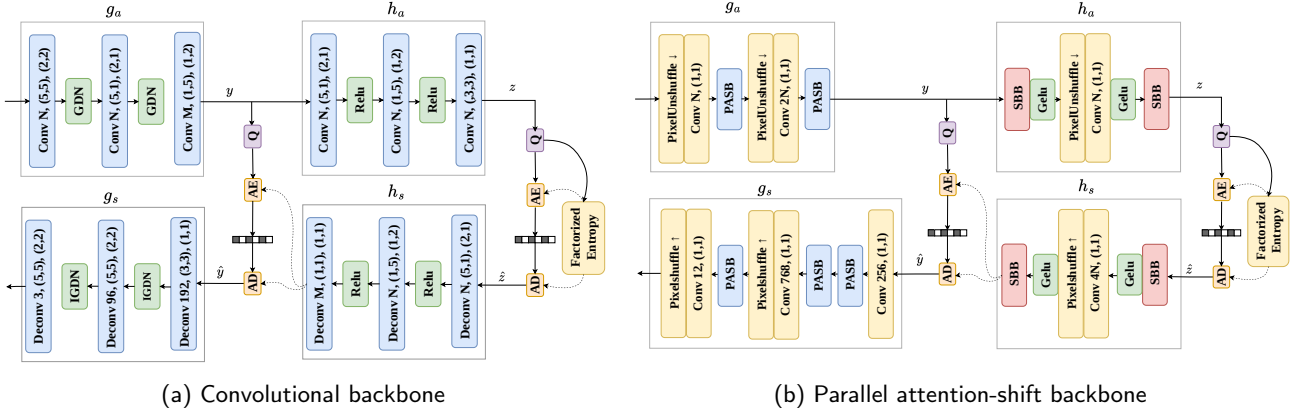


Figure 3: Architectural comparison of encoder-decoder backbones. Both configurations follow the same subband-aware processing structure, with three parallel paths for LL_2 , HF_2 , and HF_1 . (a) Baseline convolutional backbone using separable convolutions ($k \times k$ kernels with stride d factorized into $k \times 1$ (stride $d \times 1$) and $1 \times k$ (stride $1 \times d$)) with GDN/IGDN activation, where strided convolutions and transposed convolutions serve for down- and up-sampling, with channel configurations $\{N = 48, M = 96\}$ (LL_2), $\{N = 24, M = 48\}$ (HF_2), and $\{N = 16, M = 32\}$ (HF_1). (b) Proposed parallel attention-shift backbone, where PASB and SBB blocks with subband-specific LSS replace all convolutional layers, and PixelUnshuffle/Shuffle operations handle down/up-sampling. Channel dimensions are denoted in relative scale (with base N ranging up to $4N$ through PixelUnshuffle layers), where $N = 36$, $N = 24$, and $N = 16$ for LL_2 , HF_2 , and HF_1 respectively. Notations g_a , g_s , h_a , h_s , Q , AE , and AD denote the main encoder, main decoder, hyper encoder, hyper decoder, quantization, arithmetic encoding, and arithmetic decoding, respectively.

pixels. This reduces the learning burden on the encoder, allowing it to remain extremely compact. Second, the energy-concentrated subbands create a natural information-density asymmetry: the LL subband carries dense structural content, while the HF subbands are sparse. This asymmetry is precisely what enables the heterogeneous allocation of channel budgets and shift parameters across subbands, achieving a favorable rate-distortion-complexity trade-off that is not naturally available in a purely spatial-domain architecture where feature maps exhibit relatively uniform information density.

Collaborative refinement mechanism. To bridge the fidelity gap caused by the extremely lightweight encoder backbone, SWALIC introduces an asymmetric collaborative refinement scheme that operates at both ends of the compression pipeline. On the encoder side, the CBMM exploits inter-subband correlations to pre-enhance the representation of **HF** components before they are irreversibly quantized. On the decoder side, a progressive refinement pipeline is executed in two stages. First, the CRM utilizes the high-fidelity low-frequency features as spatial constraints via SFT (Wang et al., 2018) to guide high-frequency texture restoration. Second, a PostProcess Module based on global residual learning is applied after each iDWT synthesis to mitigate wavelet tiling artifacts and quantization noise. This encoder-decoder co-design keeps the encoder extremely lightweight while the decoder shoulders the bulk of the feature enhancement burden.

Entropy modeling strategy. To further align with the real-time operational requirements, SWALIC utilizes a pure hyperprior-based entropy model rather than an autoregressive context model (Ballé et al., 2018). While autoregressive models can improve R-D performance, they introduce a

serial decoding bottleneck that precludes parallel computation. By adopting a factorized hyperprior architecture, our framework ensures that all latent elements can be entropy-coded in parallel, maintaining high throughput for remote sensing data links.

Encoder-side scope and complexity definition. To provide a rigorous evaluation of the on-sensor computational burden, we define the encoding process as the complete sequence required to transform the input image into a decodable bitstream. This process includes:

- (1) the primary analysis transform g_a (incorporating the subband-aware paths and CBMM);
- (2) the hyper-analysis transform h_a ;
- (3) the hyper-synthesis transform h_s required for probability estimation.

All complexity metrics (MACs, Parameters) reported for the "Encoder" in Section 4 strictly account for the summation of these components.

3.2. Shift-based lightweight backbone design

The fundamental efficiency of the encoder and decoder critically depends on the backbone network architecture. Fig. 4 illustrates the three shift-based building blocks that replace conventional convolutional layers in our backbone. We now detail each of them in turn.

3.2.1. Learnable Spatial Shift (LSS)

A common practice in shift-based lightweight architectures is to employ a fixed grouped spatial shift: channels are partitioned into several groups, each shifted along a predetermined direction by exactly one pixel. While parameter-free, this hard assignment does not adapt to the distinct spatial statistics of different feature maps.

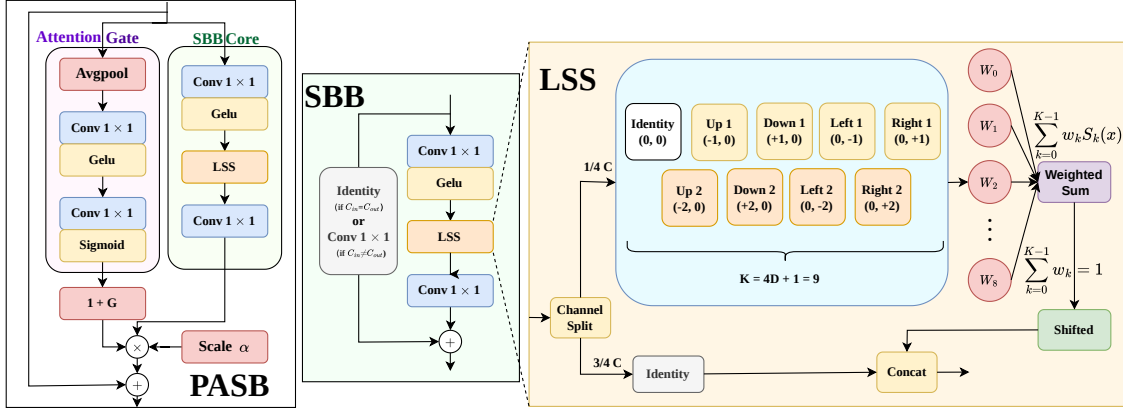


Figure 4: Architectures of the three proposed shift-based building blocks (left to right). PASB: the input is processed in parallel by an attention gating path and the SBB internal pipeline without its outer skip (SBBCore), interacting via $(1 + g)$ gating and a learnable residual scale α . SBB: a 1×1 Conv–GELU–LSS– 1×1 Conv residual block with an inverted bottleneck expand ratio E ; the skip connection is an identity when $C_{in} = C_{out}$, or a 1×1 convolution otherwise. LSS: $1/4$ of the input channels undergo a learnable weighted combination over $K = 9$ basis shift directions, while the remaining $3/4$ pass through unchanged, then both parts are concatenated.

We propose *Learnable Spatial Shift (LSS)*, a learnable soft shift operation that replaces the hard routing with a continuous weighted combination over a set of K basis directions. Formally, we define a bank of $K = 4D + 1$ base shifts consisting of an identity mapping and four cardinal directions ($\uparrow, \downarrow, \leftarrow, \rightarrow$) at D discrete distances:

$$\mathcal{S} = \{(0, 0)\} \cup \{(\pm d, 0), (0, \pm d)\}_{d=1}^D \quad (3)$$

Each basis shift $\delta_k \in \mathcal{S}$ is assigned a learnable scalar weight $w_k \in \mathbb{R}$, optimized directly via gradient descent without additional normalization:

$$\mathbf{F}_{\text{shift}} = \sum_{k=1}^K w_k \cdot \text{Roll}(\mathbf{F}, \delta_k) \quad (4)$$

Only the first $1/4$ of channels undergo shifting; the remaining $3/4$ pass through unchanged, preserving a clean identity path. This ratio is motivated by two considerations. First, in the wavelet domain, high-frequency subbands are inherently sparse—most spatial positions carry near-zero coefficients that require no spatial mixing; shifting all channels would propagate noise into these clean regions. Second, in our extremely low-dimensional encoders (16–48 channels), preserving a dominant identity path provides a stable feature manifold on which the shifted channels can operate as lightweight spatial probes, without destabilizing the overall representation. Compared to conventional fixed shift designs that partition all channels into hard-routed groups, LSS offers two distinguishing properties: (i) the continuous interpolation over multiple directions allows the network to discover an effective spatial mixing pattern through gradient-based optimization rather than relying on manual prescription; (ii) only one-fourth of the channels are shifted, reducing the risk of disrupting the original feature distribution.

Crucially, a separate *LSS* instance is maintained for each wavelet subband (LL_2, HF_2, HF_1), and within a subband it is shared across all blocks (both encoder and decoder). This subband-specific design ensures that the shift pattern for low-frequency structures differs from that for high-frequency textures, aligning with the subband-aware design philosophy of SWALIC. The shift weights are not content-adaptive—they are learned purely through gradient-based optimization during training and remain fixed at inference, making them parameter-efficient while still adaptive to subband-specific spatial statistics.

3.2.2. Shift Bottleneck Block (SBB)

Building upon the *LSS* operation, we construct *Shift Bottleneck Block (SBB)*, a lightweight residual block that adopts the 1×1 Conv–Shift– 1×1 Conv bottleneck structure (Bao et al., 2025) with two modifications tailored for our asymmetric framework.

First, we introduce an *expand ratio* E that temporarily increases the channel dimension before the shift operation (analogous to the inverted bottleneck in MobileNetV2 (Sandler et al., 2018)): the first 1×1 convolution expands C_{in} to $E \cdot C_{in}$ channels, the shift operates in this higher-dimensional space, and the second 1×1 convolution projects back to C_{out} . A larger expand ratio ($E = 2.0$) is used in the main encoder/decoder blocks where expressive power is prioritized, while $E = 1.0$ is used in the hyper encoder/decoder to minimize overhead. Second, we replace the ReLU activation with GELU (Hendrycks and Gimpel, 2016), which provides smoother gradients and improved convergence behavior (Hendrycks and Gimpel, 2016).

Formally, for an input feature $\mathbf{X} \in \mathbb{R}^{H \times W \times C_{in}}$, the block computes:

$$\text{SBB}(\mathbf{X}) = \mathbf{X} + \mathbf{W}_2(\text{Shift}(\text{GELU}(\mathbf{W}_1 \mathbf{X}))) \quad (5)$$

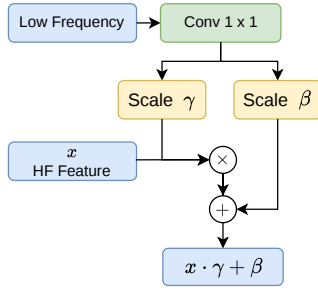


Figure 5: Detailed architecture of the Cross-Band Modulation Module (CBMM), which leverages low-frequency structural priors to guide hierarchical high-frequency feature refinement.

where $\mathbf{W}_1$ and $\mathbf{W}_2$ are 1×1 convolutions, Shift is the subband-specific LSS , and the skip connection is an identity when $C_{in} = C_{out}$ or a 1×1 convolution otherwise.

3.2.3. Parallel Attention-Shift Block (PASB)

In existing shift-based designs (Bao et al., 2025), spatial shift and channel attention are combined serially as independent components. While effective at moderate channel counts, this serial arrangement may constrain performance in our extremely lightweight encoder (e.g., HF_1 operates with as few as 16 channels): the total representational budget is split between two sequentially applied modules, and the attention and shift operations interact only indirectly.

To address this, we propose the *Parallel Attention-Shift Block (PASB)*, which fuses channel attention and spatial shift into a single parallel branch with a gated interaction mechanism. The architecture consists of two parallel paths applied to the same input $\mathbf{X} \in \mathbb{R}^{H \times W \times C}$:

- **Gating path:** A lightweight Squeeze-and-Excitation-style channel attention (Hu et al., 2018) that compresses spatial information via global average pooling, then produces per-channel gating coefficients through a two-layer bottleneck ($C \rightarrow C/4 \rightarrow C$) with sigmoid activation: $\mathbf{g} = \sigma(\mathbf{W}_g^2 \cdot \text{GELU}(\mathbf{W}_g^1 \cdot \text{AvgPool}(\mathbf{X}))) \in (0, 1)^C$.
- **Spatial path:** The SBB transformation pipeline without the skip connection: $\mathbf{S} = \mathbf{W}_2(\text{Shift}(\text{GELU}(\mathbf{W}_1 \mathbf{X})))$. The SBB’s internal skip is deliberately removed here: the PASB’s outer residual connection ($\mathbf{X} + \dots$) already preserves the identity signal, and including $\mathbf{X}$ inside the spatial path would entangle the clean input with the gate-modulated branch, undermining the $(1 + \mathbf{g})$ structure’s role as a pure shift-feature amplifier.

The two paths interact through a gated enhancement mechanism:

$$\text{PASB}(\mathbf{X}) = \mathbf{X} + \alpha \cdot (\mathbf{S} \odot (1 + \mathbf{g})) \quad (6)$$

where $\odot$ denotes channel-wise multiplication, and α is a learnable residual scale parameter initialized to 0.1.

Three aspects of this formulation are noteworthy. **First**, the $(1 + \mathbf{g})$ gating structure (rather than the conventional $\mathbf{g}$) ensures that the gating coefficients lie in $(1, 2)$ instead of $(0, 1)$. This guarantees that the attention mechanism can only *amplify* informative channels, never suppress them—a critical safeguard in our extremely low-dimensional latent space (16–48 channels) where any channel suppression risks irreversible information loss. **Second**, the learnable scale α acts as a training stabilizer: initialized to 0.1, it forces the block to approximate an identity mapping in early training, allowing the spatial and attention branches to be gradually activated as training progresses. **Third**, by placing the attention gate and the spatial shift in the *same* parallel branch, the gating coefficients directly modulate the output of the shift operation. This “gated shift” interaction enables the attention to selectively amplify shift features on a per-channel basis, creating a tighter coupling than conventional serial arrangements where attention and shift operate independently. Additionally, the parallel topology shortens gradient paths during backpropagation: error signals reach both branches directly without attenuation through intermediate layers, promoting stable convergence even at low learning rates.

In the hyper encoder/decoder, where computational cost must be minimized, we use standalone *SBB* blocks. *PASB* blocks are deployed in the main encoder (g_a) and decoder (g_s) where their enhanced representational capacity provides the greatest benefit.

3.3. Cross-band modulation module (CBMM)

To further enhance feature representation under stringent complexity constraints, we propose the CBMM at the encoder side. Unlike conventional LIC methods that treat subbands as independent entities, CBMM explicitly leverages the inherent *spatial-structural correlations* across different wavelet decomposition levels (Shapiro, 1993).

Specifically, for each decomposition level $l \in \{1, 2\}$, the low-frequency subband feature $\mathbf{F}_{LL,l} \in \mathbb{R}^{H_l \times W_l \times C_l}$ serves as a spatial reference. Let $\mathbf{F}_{HF,l} \in \mathbb{R}^{H_l \times W_l \times 3C_l}$ denote the concatenated high-frequency features $\{\mathbf{F}_{LH,l}, \mathbf{F}_{HL,l}, \mathbf{F}_{HH,l}\}$. The module generates spatially-adaptive modulation parameters—scale ($\gamma_{e,l}$) and shift ($\beta_{e,l}$)—through an efficient pointwise convolution layer ϕ_l :

$$\gamma_{e,l}, \beta_{e,l} = \phi_l(\mathbf{F}_{LL,l}) \quad (7)$$

where ϕ_l is implemented as a 1×1 convolution layer that maps C_l input channels to $6C_l$ output channels. The modulated high-frequency feature $\mathbf{F}'_{HF,l}$ is then obtained via a spatial-wise affine transformation:

$$\mathbf{F}'_{HF,l} = \mathbf{F}_{HF,l} \odot \gamma_{e,l} + \beta_{e,l} \quad (8)$$

where $\odot$ denotes element-wise multiplication. The rationale for employing 1×1 pointwise convolutions is twofold:

- (1) it minimizes the MACs count, which is critical for sensor-side deployment;
- (2) it effectively captures the cross-channel correlations between the structural backbone (LL) and texture residuals (HF).

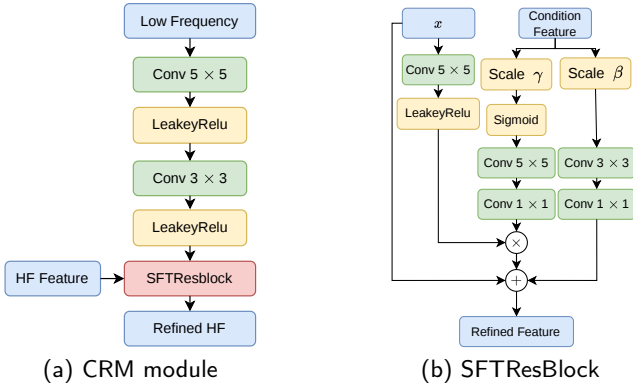


Figure 6: Detailed architectural components of the CRM. (a) Contextual Refinement Module (CRM) for cross-band feature restoration. (b) Internal structure of the SFTResBlock, which utilizes structural priors for spatial modulation.

This allows the encoder to "pre-enhance" high-frequency representations under global structural guidance, effectively preserving critical details before the irreversible information loss incurred by quantization.

Compared with deepening the HF encoder via ordinary ResBlocks, CBMM offers two advantages. First, a plain ResBlock is confined to its own subband and has no mechanism for cross-band information exchange. After DWT decomposition, the LL and HF subbands are inherently correlated—a strong edge in LL typically coincides with textured regions in HF at the same spatial location. CBMM exploits this dependency by generating spatially adaptive modulation parameters from the LL features through a single 1×1 convolution, effectively allowing the structural backbone to guide HF feature extraction. Second, this cross-band communication is achieved with negligible cost (0.002 M parameters, 0.001 GMACs), whereas achieving a comparable receptive field through additional ResBlocks would require substantially more parameters and computation without establishing the same cross-band information flow.

3.4. Contextual refinement module (CRM)

The CRM is the cornerstone of our asymmetric decoder, designed to restore lost textures by exploiting inter-frequency dependencies. While the encoder-side CBMM employs efficient 1×1 convolutions to minimize overhead, the CRM at the ground station leverages the superior computational capacity to perform a more sophisticated SFT (Wang et al., 2018).

The CRM employs a condition network ψ_l with a 5×5 convolution kernel to capture a broader spatial context from the reconstructed low-frequency subband $\hat{\mathbf{F}}_{LL,l}$. Unlike the 1×1 kernels used in the encoder's CBMM, the larger 5×5 kernels here allow the decoder to recover fine textures by leveraging structural priors from a wider spatial context of $\hat{\mathbf{F}}_{LL,l}$. This produces dense modulation parameters $\gamma_{d,l}$ and $\beta_{d,l}$:

$$\gamma_{d,l}, \beta_{d,l} = \psi_l(\hat{\mathbf{F}}_{LL,l}) \quad (9)$$

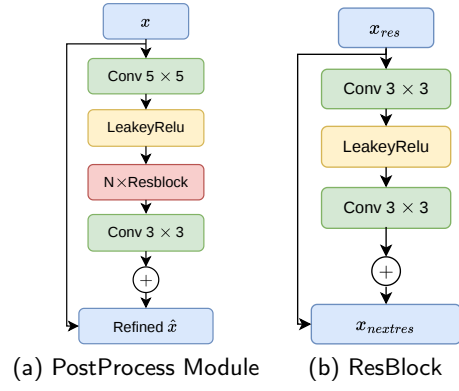


Figure 7: Detailed architectural components of the Post-Process Module. (a) PostProcess Module for spatial domain artifact mitigation. (b) Residual Block (ResBlock) architecture employed in the global residual learning-based postprocessing pipeline.

To ensure training stability and preserve identity features, the SFT is embedded within a Residual Block structure (SFT-ResBlock). For a given input feature $\mathbf{F}_{in}$ derived from the quantized high-frequency stream, the refined output $\hat{\mathbf{F}}_{HF,l}^{ref}$ is computed as:

$$\mathbf{F}_{mid} = \sigma(\text{Conv}_1(\mathbf{F}_{in})) \odot \gamma_{d,l} + \beta_{d,l} \quad (10)$$

$$\hat{\mathbf{F}}_{HF,l}^{ref} = \mathbf{F}_{in} + \text{Conv}_2(\mathbf{F}_{mid}) \quad (11)$$

where $\sigma(\cdot)$ denotes the LeakyReLU activation and $\text{Conv}_1, \text{Conv}_2$ represent 3×3 convolutional layers.

Compared with a deeper stack of ordinary ResBlocks at the decoder, CRM is more effective for two reasons. First, standard ResBlocks operate exclusively on their own quantized and degraded HF features, whereas at the decoder the reconstructed LL features already provide a high-quality structural scaffold. CRM leverages this external guidance via the SFT mechanism, which has been shown in image super-resolution to be effective at recovering realistic textures from structural priors (Wang et al., 2018). Second, the 5×5 convolutional condition network provides a broader spatial context than the typical 3×3 kernels of ResBlocks, allowing the modulation parameters to capture larger-scale structural cues. This external guidance is not available to a ResBlock-only design.

3.5. Postprocess module

After the iDWT synthesis, the reconstructed image components are further refined through a multi-stage PostProcess Module. To align with the hierarchical nature of the two-stage wavelet decomposition, we implement a stage-adaptive refinement strategy where the depth of the residual network is tailored to the spatial resolution and information density of each reconstruction level.

The module is optimized via *global residual learning* (Zhang et al., 2017), where the network predicts a residual

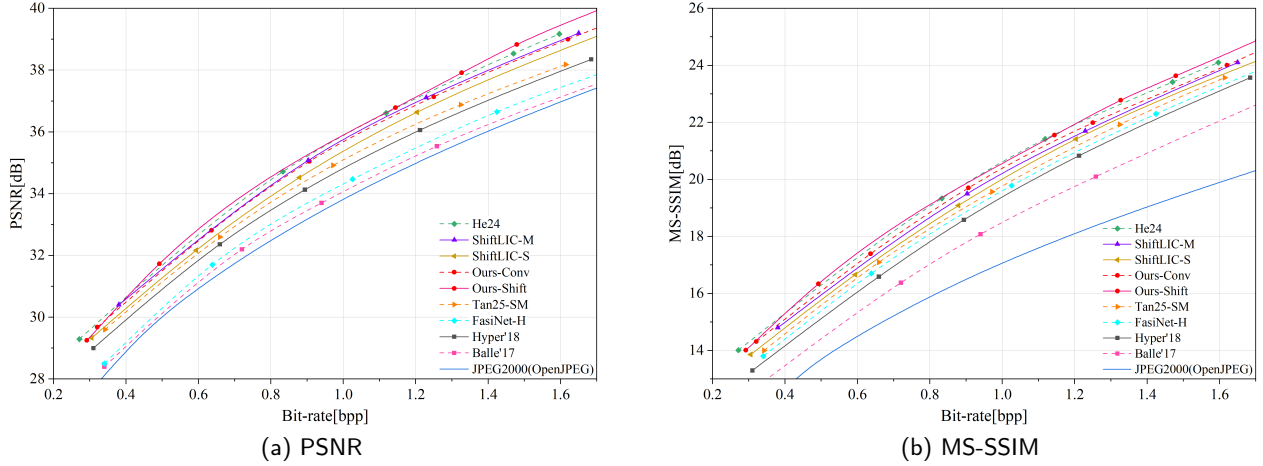


Figure 8: Rate-distortion (R-D) performance comparison on the UCM dataset. Ours-Conv and Ours-Shift denote the convolutional and parallel attention-shift backbones, respectively, each equipped with the full refinement pipeline (CBMM + CRM + PostProcess Module). MS-SSIM values are represented in decibels (dB), calculated as $-10 \log_{10}(1 - s)$, where $s \in [0, 1]$ denotes the original MS-SSIM index.

mapping $\mathcal{R}(\cdot)$ to compensate for quantization noise and synthesis artifacts. Specifically, for each reconstruction stage, the refined output is obtained by:

$$\mathbf{F}_{refined} = \mathbf{F}_{coarse} + \mathcal{R}(\mathbf{F}_{coarse}) \quad (12)$$

where $\mathbf{F}_{coarse}$ denotes the initial synthesis result from the iDWT. In our implementation, the complexity of the residual learning is governed by the number of stacked Residual Blocks, denoted as N .

Stage 1: LL_2 to LL_1 reconstruction. During the synthesis of the first-level low-frequency subband $\hat{\mathbf{L}}_1$ from $\hat{\mathbf{L}}_2$ and the second-level high-frequency components, we set $N = 2$. At this lower resolution (1/4 of the original size), the primary objective is to stabilize the structural backbone. A shallower network is sufficient for this stage, as the information density is relatively concentrated and the spatial complexity is limited.

Stage 2: LL_1 to spatial reconstruction. During the final synthesis from $\hat{\mathbf{L}}_1$ to the spatial-domain image $\hat{\mathbf{x}}$, the complexity is increased to $N = 3$. The additional depth allows the module to better handle the significantly increased spatial complexity and finer textures inherent in the full-resolution representation. By employing a 5×5 "head" convolution to expand the receptive field followed by this deeper residual chain, the module effectively restores high-frequency fidelity that was constrained at the encoder side.

This hierarchical design ensures that the most intensive refinement occurs at the final stage to maximize visual fidelity, while the overall decoder complexity remains manageable for server-side processing.

4. Experiments

4.1. Experiments details

Training. We implemented SWALIC using the PyTorch-based CompressAI library (Bégaint et al., 2020). The AID

dataset (Xia et al., 2017), which contains diverse aerial scenes, was used for training. All training images were randomly cropped into patches of size 256×256 . The models were optimized using the Adam optimizer (Kingma and Ba, 2015) with a batch size of 32 and were trained for 1000 epochs. The initial learning rate was set to 2×10^{-3} , the patience parameter was 30, and the reduction factor was 0.5.

The joint optimization objective is defined as $\mathcal{L} = R + \lambda \cdot D$, where the distortion term D represents the Mean Squared Error (MSE). To ensure a rigorous and unbiased comparison, all LIC methods were retrained from scratch on the AID dataset to align the data distribution. For MSE-optimized models, the Lagrange multiplier λ was selected from the set $\{0.005, 0.01, 0.05, 0.075, 0.1, 0.15\}$.

Testing. We evaluated the models on the UC Merced (UCM) Land Use dataset (Yang and Newsam, 2010), a standard benchmark in remote sensing consisting of 21 land-cover categories. The test set comprises images with a resolution of 256×256 .

Performance was quantified using Peak Signal-to-Noise Ratio (PSNR) and Multi-Scale Structural Similarity (MS-SSIM) (Wang et al., 2003) for reconstruction quality, and bits per pixel (BPP) for bit rate. To assess real-world deployability on edge hardware, encoding and decoding latencies were benchmarked on an Intel Xeon E5-2678 v3 CPU, representing a standard non-GPU execution environment.

Rate-Distortion performance. We compare SWALIC against several state-of-the-art lightweight LIC models, including ShiftLIC-M/S (Bao et al., 2025), FasiNet-H (Zheng et al., 2021), Tan25-SM (Tan et al., 2025), and the traditional JPEG2000 standard. As illustrated by the R-D curves in Fig. 8, both our convolutional baseline (Ours-Conv) and shift-based model (Ours-Shift) achieve competitive performance across a wide range of bitrates, with Ours-Shift

Table 1

Quantitative evaluation of complexity and latency. Measured with an input size of 256×256 . Latency is measured on an Intel Xeon E5-2678 V3 CPU. Bold indicates best performance in each metric.

Model	Params (M)		MACs (G)		Latency (ms)	
	Enc.T	Dec.T	Enc.T	Dec.T	Enc.T	Dec.T
He24	9.39	2.41	8.61	10.59	845	1436
ShiftLIC-M	2.25	3.54	5.60	5.79	286	286
ShiftLIC-S	2.63	1.34	3.84	3.66	220	236
Tan25-SM	0.94	0.58	1.10	1.08	279	271
FasiNet-H	1.62	0.278	0.73	1.26	77	71
Hyper'18	3.65	1.46	2.70	9.66	149	190
Ballé'17	1.46	1.46	2.42	9.66	98	157
Ours-Shift	0.119	2.931	0.044	6.55	52	323

consistently outperforming Ours-Conv—validating the efficacy of replacing heavy convolutions with memory-efficient spatial shift operations.

While more complex methods such as He24 (He et al., 2024) achieve R-D performance comparable to Ours-Shift, this comes at the cost of substantially higher encoder overhead—He24's encoder, for instance, requires 9.39 M parameters and 8.61 GMACs, roughly two orders of magnitude more than Ours-Shift (0.119 M, 0.044 G). Ours-Shift thus achieves essentially the same reconstruction fidelity at a fraction of the encoder cost, confirming the effectiveness of the asymmetric design strategy.

Complexity and latency analysis. We further analyze the performance-complexity trade-off to highlight the encoder's efficiency. As summarized in Table 1, Ours-Shift requires only 0.119 M parameters and 0.044 GMACs at the encoder—representing a 95.5% reduction in parameter count and a 98.9% reduction in MACs compared to ShiftLIC-S (Bao et al., 2025). It also achieves the lowest encoding latency at 52 ms on the same Intel Xeon CPU used for all benchmarks, a 32.5% improvement over FasiNet-H (Zheng et al., 2021). These results confirm that the asymmetric design of SWALIC effectively concentrates computational burden at the decoder, where resources are abundant, while keeping the encoder footprint minimal for on-sensor deployment.

Latency and memory at ultra-high resolution. To evaluate deployability under extreme workloads, we measure latency and peak CPU memory on 3840×3840 images using the same Intel Xeon E5-2678 V3 CPU. As shown in Table 2, Ours-Shift achieves the lowest encoder latency (7.0 s) and the lowest memory footprint (~ 115 MB), substantially lower than ShiftLIC (62–89 s, 284 MB) and FasiNet-H (21 s, 685 MB). Comparing Ours-Conv and Ours-Shift, the encoder-side latency and memory remain nearly unchanged (6.7 s $\rightarrow$ 7.0 s, 115.0 MB $\rightarrow$ 115.3 MB), indicating that the shift-based backbone introduces negligible encoder overhead. The decoder latency increases moderately (78.3 s $\rightarrow$ 93.7 s), which is acceptable since the decoder

Table 2

Latency (seconds) and peak CPU memory (MB) on 3840×3840 images.

Model	Latency (s)		Memory (MB)	
	Enc.T	Dec.T	Enc.T	Dec.T
FasiNet-H	21.1	19.2	684.5	677.1
ShiftLIC-M	88.7	87.1	283.9	283.4
ShiftLIC-S	62.1	63.1	283.9	283.4
Tan25-SM	89.2	87.0	171.8	170.9
He24	233.2	256.1	169.0	118.2
Ballé'17	24.2	38.3	178.9	172.4
Hyper'18	27.9	39.6	175.1	174.4
Ours-Conv	6.7	78.3	115.0	114.6
Ours-Shift	7.0	93.7	115.3	114.6

operates on resource-rich ground stations. This trade-off is justified by the 4.09% BD-Rate improvement gained from the shift backbone (Table 4). Overall, SWALIC's wavelet-domain decomposition and subband-aware design deliver robust efficiency at ultra-high resolutions without compromising the encoder's minimal resource profile.

4.2. Ablation study

We conduct two complementary ablation studies on the UCM dataset. All ablation experiments are performed on the convolutional backbone, which provides a simpler and more interpretable baseline for isolating the contribution of each refinement module without the confounding interactions introduced by the shift-based operations. We first evaluate the collaborative refinement modules (CBMM, CRM, and Post-Process Module) on the convolutional backbone (Table 3), then assess the gain of upgrading to the parallel attention-shift backbone on top of the best refinement configuration (Table 4).

Effectiveness of individual modules. Each module provides a meaningful gain when deployed alone. CBMM (Case 2 vs. Case 1) improves BD-Rate by 4.35%, CRM (Case 3) by 9.30%, and PostProcess Module (Case 4) by 12.74%. Notably, CBMM achieves this with negligible encoder-side overhead (0.002 M parameters, 0.001 GMACs), confirming that cross-band modulation before quantization is a cost-effective strategy for preserving structural information.

Collaborative design of CBMM and decoder modules. A closer inspection of two-module combinations reveals that CBMM does not always combine additively with a single decoder module. While CRM+PostProcess Module (Case 7) reduces BD-Rate to +1.14%, adding CBMM to CRM alone (Case 5, +7.53%) is worse than CRM alone (+6.75%), and adding CBMM to PostProcess Module alone (Case 6, +4.91%) is worse than PostProcess Module alone (+3.31%). This pattern indicates that CBMM's benefit is contingent on the presence of both CRM and PostProcess Module: when only one decoder module is active, the encoder-side pre-enhancement does not translate into a net gain. When both CRM and PostProcess Module are

Table 3

Ablation studies of the collaborative refinement modules. All cases are built upon the convolutional backbone. The anchor for BD-Rate (%) is Case 8 (full configuration). MACs are measured with an input size of 256×256 . Bold indicates the full configuration.

Method	CBMM	CRM	Post	BD-Rate (%) ↓	Params (M)		MACs (G)	
					Enc.T	Dec.T	Enc.T	Dec.T
Case 1	✗	✗	✗	+16.05	0.164	1.898	0.062	3.44
Case 2	✓	✗	✗	+11.70	0.166	1.898	0.063	3.44
Case 3	✗	✓	✗	+6.75	0.164	1.914	0.062	3.60
Case 4	✗	✗	✓	+3.31	0.164	1.998	0.062	7.96
Case 5	✓	✓	✗	+7.53	0.166	1.914	0.063	3.60
Case 6	✓	✗	✓	+4.91	0.166	1.998	0.063	7.96
Case 7	✗	✓	✓	+1.14	0.164	2.014	0.062	8.12
Case 8	✓	✓	✓	Anchor(0%)	0.166	2.014	0.063	8.12

Table 4

Backbone architecture ablation. Both configurations employ the full refinement pipeline (CBMM + CRM + PostProcess Module), and the convolutional backbone serves as the BD-Rate anchor.

Backbone	BD-Rate (%) ↓	Params (M)		MACs (G)	
		Enc.T	Dec.T	Enc.T	Dec.T
Ours-Conv	Anchor (0%)	0.166	2.014	0.063	8.12
Ours-Shift	-4.09	0.119	2.931	0.044	6.55

present (Case 7 → Case 8), however, CBMM contributes an additional 1.14% BD-Rate reduction, reaching the overall optimum. This supports the view that CBMM and the progressive decoder pipeline are co-designed: the CRM first reconstructs HF features using LL structural priors via SFT, and the PostProcess Module then refines the spatial-domain output through global residual learning; only this complete pipeline fully unlocks the benefit of encoder-side cross-band modulation.

Complexity stability. As shown in Table 3, encoder-side complexity remains nearly constant at 0.062–0.063 GMACs across all configurations, regardless of which refinement modules are activated. The additional computation introduced by CRM and PostProcess Module is entirely borne by the decoder, strictly preserving the asymmetric design philosophy of SWALIC.

Effectiveness of parallel attention-shift backbone. To isolate the contribution of the proposed parallel attention-shift backbone, we compare two configurations—both employing the full refinement pipeline (CBMM + CRM + PostProcess Module) but differing only in backbone architecture. As shown in Table 4, replacing the separable convolutional backbone with the parallel attention-shift design reduces the BD-Rate by 4.09%, while simultaneously decreasing encoder parameters by 28.3% (0.166 → 0.119 M) and encoder MACs by 30.2% (0.063 → 0.044 G). The decoder parameter count increases moderately (2.014 → 2.931 M) due to the deeper stack of PASB blocks, while decoder MACs decrease (8.12 → 6.55 G) owing to the

computational efficiency of shift operations replacing heavy convolutions. Both trends are consistent with the asymmetric design philosophy: the decoder has ample capacity to absorb additional parameters, and the MACs reduction benefits overall throughput. Taken together, the parallel attention-shift backbone improves both compression efficiency and encoder-side resource economy—a dual benefit critical for SWaP-constrained deployment.

4.3. Qualitative visualization

To evaluate the perceptual quality of the reconstructed images, we provide a qualitative comparison between SWALIC variants and the traditional JPEG2000 standard on the UCM dataset. Fig. 9 illustrates the visual results of an airplane scene at approximately 0.09 ~ 0.10 bpp, where the variants follow the same naming and configuration as defined in the caption.

As observed in the close-up patches, JPEG2000 suffers from severe information loss and noticeable color artifacts, completely distorting the high-contrast boundaries of the aircraft fuselage. These distortions stem from the fixed transform kernels of traditional wavelets, which struggle to preserve sharp structural primitives at low bitrates.

The progressive improvement across the configurations demonstrates the effectiveness of both the proposed refinement modules and the backbone architecture. Starting from Ours-Base, the image eliminates the blurring and color artifacts present in JPEG2000, but is severely over-smoothed: the blue livery stripes on the fuselage are almost entirely lost, blurred into a uniform grey patch. By integrating the CRM and PostProcess Module at the decoder side, Ours-V1 mitigates the overall blurring and produces sharper wing contours with enhanced three-dimensionality; however, high-frequency textures such as the blue livery remain largely unrecovered. The subsequent addition of the encoder-side CBMM in Ours-V2 brings cross-band modulation into effect: the outlines of the dark fuselage stripes begin to emerge and the overall sharpness improves noticeably.

Most notably, the final Ours-Full configuration achieves a clear improvement. By replacing the conventional convolution-based backbone with the proposed shift-based PASB/SBB

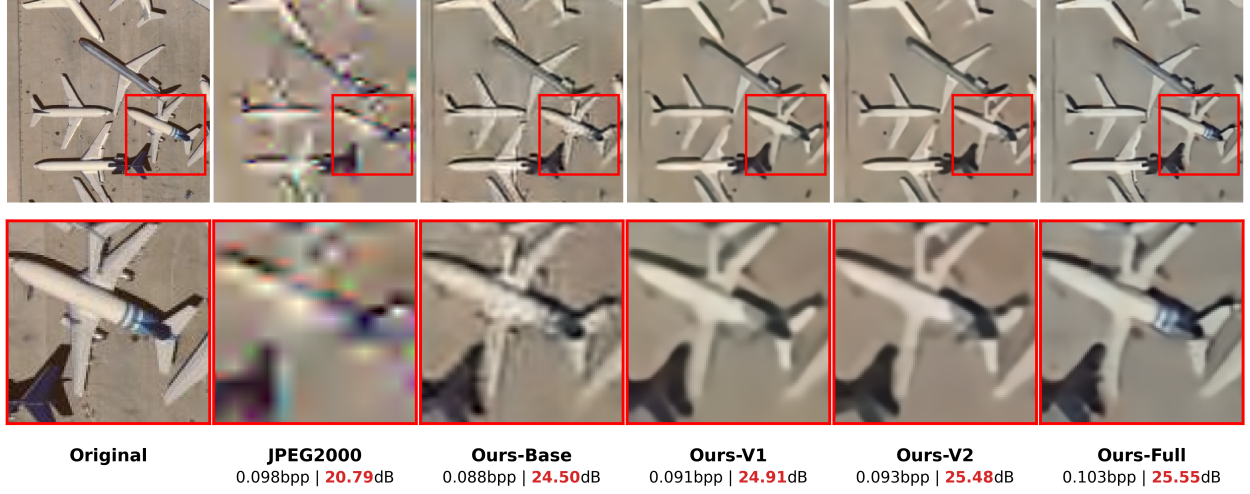


Figure 9: Qualitative comparison on the UCM dataset. Reconstructions and close-ups of an airplane scene encoded by JPEG2000 and SWALIC variants at approximately 0.09 ~ 0.10 bpp. Variants are defined as: Ours-Base (convolutional backbone, no refinement), Ours-V1 (convolutional backbone + CRM + PostProcess Module), Ours-V2 (convolutional backbone + CBMM + CRM + PostProcess Module), and Ours-Full (shift-based backbone + CBMM + CRM + PostProcess Module). Annotations indicate bpp and PSNR (dB). The progressive refinement from Ours-Base to Ours-Full demonstrates the complementary contributions of both the collaborative refinement modules and the shift-based backbone architecture.

architecture, it not only reduces encoder parameters and MACs but also better preserves high-frequency structural integrity. As shown in the enlarged regions, Ours-Full successfully reconstructs the distinct blue and white livery banding on the fuselage and the sharp shadows along the trailing edge of the wings—details that were completely absent in the baseline versions.

These visual results demonstrate that our asymmetric scheme successfully recovers high-frequency details at the decoder side. Despite the encoder being extremely lightweight (0.119 M parameters), the collaborative synergy between the shift-based subband processing and the spatial-domain refinement enables SWALIC to maintain high-fidelity reconstruction suitable for real-time deployment on SWaP-constrained edge platforms.

5. Conclusion

In this work, we proposed SWALIC, a wavelet-based lightweight LIC framework specifically optimized for asymmetric sensor-server ecosystems. By integrating two-stage biorthogonal wavelet decomposition with a subband-aware complexity allocation strategy and a parallel attention-shift backbone, the framework achieves an extremely efficient encoding process, requiring only 0.119 M parameters and 0.044 GMACs. To bridge the fidelity gap inherent in lightweight encoding, we introduced an end-to-end collaborative refinement scheme. Whereas the encoder-side CBMM pre-enhances high-frequency features under structural guidance with negligible overhead, the decoder-side CRM and PostProcess Module effectively utilize cross-band structural priors and global residual learning to restore fine textures and mitigate quantization artifacts. The parallel attention-shift backbone, built on learnable spatial shift

operations, reduces encoder complexity by 28.3% (0.166 → 0.119 M parameters) and 30.2% (0.063 → 0.044 GMACs) over its convolutional counterpart while improving BD-Rate by 4.09%, demonstrating that shift operations are a compelling alternative to heavy convolutions in asymmetric LIC encoders.

Experimental results on the UCM dataset demonstrate that SWALIC delivers rate-distortion performance competitive with advanced lightweight methods while reducing encoder-side computational demands by over 90% compared to existing benchmarks. Latency evaluations on a commodity Intel Xeon CPU further confirm its suitability for real-time deployment on SWaP-constrained edge platforms, such as UAV cameras and satellite imaging sensors. We note that the current evaluation is confined to remote sensing benchmarks (AID and UCM), consistent with our target application of satellite and UAV imaging. Extending the evaluation to natural-image datasets such as Kodak and CLIC, as well as exploring more adaptive wavelet-like transforms, are important directions for future work.

6. ACKNOWLEDGEMENT

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References

- Mohsan, S.A.H., Khan, M.A., Noor, F., Ullah, I., Alsharif, M.H., 2022. Towards the unmanned aerial vehicles (UAVs): A comprehensive review. *Drones* 6, 147.

- Yang, W., Wang, M., Zhang, J., Zou, J., Hua, M., Xia, T., You, X., 2017. Narrowband wireless access for low-power massive Internet of Things: A bandwidth perspective. *IEEE Wireless Commun.* 24, 138–145.
- Chintalapati, B., Precht, A., Hanra, S., Laufer, R., Liwicki, M., Eickhoff, J., 2025. Opportunities and challenges of on-board AI-based image recognition for small satellite Earth observation missions. *Adv. Space Res.* 75, 6734–6751.
- Sang, H., Zhang, L., Chen, T., Guo, W., Zhang, Z., 2026. Onboard deployment of remote sensing foundation models: A comprehensive review of architecture, optimization, and hardware. *Remote Sens.* 18, 298.
- Wang, F., Zhang, M., Wang, X., Ma, X., Liu, J., 2020. Deep learning for edge computing applications: A state-of-the-art survey. *IEEE Access* 8, 58322–58336.
- Bao, Y., Tan, W., Jia, C., Li, M., Liang, Y., Tian, Y., 2025. ShiftLIC: Lightweight learned image compression with spatial-channel shift operations. *IEEE Trans. Circuits Syst. Video Technol.* .
- Mishra, D., Singh, S.K., Singh, R.K., 2021. Wavelet-based deep auto encoder-decoder (WDAED)-based image compression. *IEEE Trans. Circuits Syst. Video Technol.* 31, 1452–1462.
- Zhang, Y., Lin, K., Jia, C., Ma, S., 2022. Interpretable learned image compression: A frequency transform decomposition perspective, in: *Proc. Data Compression Conf. (DCC)*, p. 495.
- Ma, H., Liu, D., Xiong, R., Wu, F., 2020. iWave: CNN-based wavelet-like transform for image compression. *IEEE Trans. Multimedia* 22, 1667–1679.
- He, Z., Huang, M., Luo, L., Yang, X., Zhu, C., 2024. Towards real-time practical image compression with lightweight attention. *Expert Syst. Appl.* 252, 124142.
- Tan, W., Bao, Y., Meng, F., Liang, Y., 2025. Grouped transform for ultra-low-complexity learned image compression, in: *Proc. IEEE Int. Symp. Circuits Syst. (ISCAS)*, pp. 1–5.
- Zheng, Z., Wang, X., Lin, X., Lv, S., 2021. Get the best of the three worlds: Real-time neural image compression in a non-GPU environment, in: *Proc. 29th ACM Int. Conf. Multimedia*, pp. 5400–5409.
- Wallace, G.K., 1991. The jpeg still picture compression standard. *Commun. ACM* 34, 30–44.
- Rabbani, M., Joshi, R., 2002. An overview of the jpeg 2000 still image compression standard. *Signal Process.: Image Commun.* 17, 3–48.
- Antonini, M., Barlaud, M., Mathieu, P., Daubechies, I., 1992. Image coding using wavelet transform. *IEEE Trans. Image Process.* 1, 205–220.
- Tausif, M., Jain, A., Khan, E., Hasan, M., 2021. Memory-efficient architecture for FrWF-based DWT of high-resolution images for IoMT applications. *Multimedia Tools Appl.* 80, 11177–11199.
- Ballé, J., Laparra, V., Simoncelli, E.P., 2017. End-to-end optimized image compression, in: *Proc. Int. Conf. Learn. Represent. (ICLR)*.
- Ballé, J., Minnen, D., Singh, S., Hwang, S.J., Johnston, N., 2018. Variational image compression with a scale hyperprior, in: *Proc. Int. Conf. Learn. Represent. (ICLR)*.
- Minnen, D., Ballé, J., Toderici, G., 2018. Joint autoregressive and hierarchical priors for learned image compression, in: *Advances in Neural Information Processing Systems (NeurIPS)*.
- He, D., Zheng, Y., Sun, B., Wang, Y., Qin, H., 2021. Checkerboard context model for efficient learned image compression, in: *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, pp. 14771–14780.
- Minnen, D., Singh, S., 2020. Channel-wise autoregressive entropy models for learned image compression, in: *Proc. IEEE Int. Conf. Image Process. (ICIP)*, pp. 3339–3343.
- Cheng, Z., Sun, H., Takeuchi, M., Katto, J., 2020. Learned image compression with discretized gaussian mixture likelihoods and attention modules, in: *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, pp. 7939–7948.
- He, D., Zheng, Y., Sun, B., Wang, Y., Qin, H., 2022. Elic: Efficient learned image compression with unevenly grouped space-channel contextual adaptive coding, in: *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, pp. 5718–5727.
- Ma, H., Liu, D., Yan, N., Li, H., Wu, F., 2022. End-to-end optimized versatile image compression with wavelet-like transform. *IEEE Trans. Pattern Anal. Mach. Intell.* 44, 1247–1263.
- Rippel, O., Bourdev, L., 2019. Real-time adaptive image compression, in: *Proc. Int. Conf. Mach. Learn. (ICML)*.
- Luo, A., Sun, H., Liu, J., Katto, J., 2022. Memory-efficient learned image compression with pruned hyperprior module, in: *Proc. IEEE Int. Conf. Image Process. (ICIP)*, pp. 3061–3065.
- Kim, J.H., Choi, J.H., Chang, J., Lee, J.S., 2020. Efficient deep learning-based lossy image compression via asymmetric autoencoder and pruning, in: *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, pp. 2063–2067.
- Tan, W., Bao, Y., Meng, F., Li, C., Zhu, L., Liang, Y., 2025. Structured sparsity learning for efficient learned image compression, in: *Proc. IEEE Int. Symp. Circuits Syst. (ISCAS)*, pp. 1–5.
- Sun, H., Yu, L., Katto, J., 2025. Q-LIC: Quantizing learned image compression with channel splitting. *IEEE Trans. Circuits Syst. Video Technol.* 35, 3798–3811.
- Chen, Y., Lyu, Z., He, B., Cao, N., Chen, G., Lu, G., Zhang, W., 2025. Knowledge distillation for learned image compression, in: *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, pp. 4996–5006.
- Liu, D., Zhang, J., Liang, X., Qi, Y., Song, Y., Xi, Y., Jin, J., 2026. RS-LLIC: A lightweight learned image compression model with knowledge distillation for onboard remote sensing. *IEEE Trans. Geosci. Remote Sens.* 64, 1–13.
- Cohen, A., Daubechies, I., Feauveau, J.C., 1992. Biorthogonal bases of compactly supported wavelets. *Commun. Pure Appl. Math.* 45, 485–560.
- Wang, X., Yu, K., Dong, C., Loy, C.C., 2018. Recovering realistic texture in image super-resolution by deep spatial feature transform, in: *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, pp. 606–615.
- Sandler, M., Howard, A., Zhu, M., Zhmoginov, A., Chen, L.C., 2018. MobileNetV2: Inverted residuals and linear bottlenecks, in: *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, pp. 4510–4520.
- Hendrycks, D., Gimpel, K., 2016. Gaussian error linear units (GELUs). *arXiv preprint arXiv:1606.08415* .
- Hu, J., Shen, L., Sun, G., 2018. Squeeze-and-excitation networks, in: *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, pp. 7132–7141.
- Shapiro, J.M., 1993. Embedded image coding using zerotrees of wavelet coefficients. *IEEE Trans. Signal Process.* 41, 3445–3462.
- Zhang, K., Zuo, W., Chen, Y., Meng, D., Zhang, L., 2017. Beyond a gaussian denoiser: Residual learning of deep cnn for image denoising. *IEEE Trans. Image Process.* 26, 3142–3155.
- Bégaint, J., Racapé, F., Feltman, S., Pushparaja, A., 2020. Compressai: A PyTorch library and evaluation platform for end-to-end compression research. *ArXiv:2011.03029*.
- Xia, G.S., Hu, J., Hu, F., Shi, B., Bai, X., Zhong, Y., Zhang, L., 2017. Aid: A benchmark data set for performance evaluation of aerial scene classification. *IEEE Trans. Geosci. Remote Sens.* 55, 3965–3981.
- Kingma, D.P., Ba, J., 2015. Adam: A method for stochastic optimization, in: *Proc. Int. Conf. Learn. Represent. (ICLR)*.
- Yang, Y., Newsam, S., 2010. Bag-of-visual-words and spatial extensions for land-use classification, in: *Proc. 18th SIGSPATIAL Int. Conf. Adv. Geographic Inf. Syst.*, pp. 270–279.
- Wang, Z., Simoncelli, E.P., Bovik, A.C., 2003. Multiscale structural similarity for image quality assessment, in: *Proc. 37th Asilomar Conf. Signals, Syst., Comput.*, pp. 1398–1402.